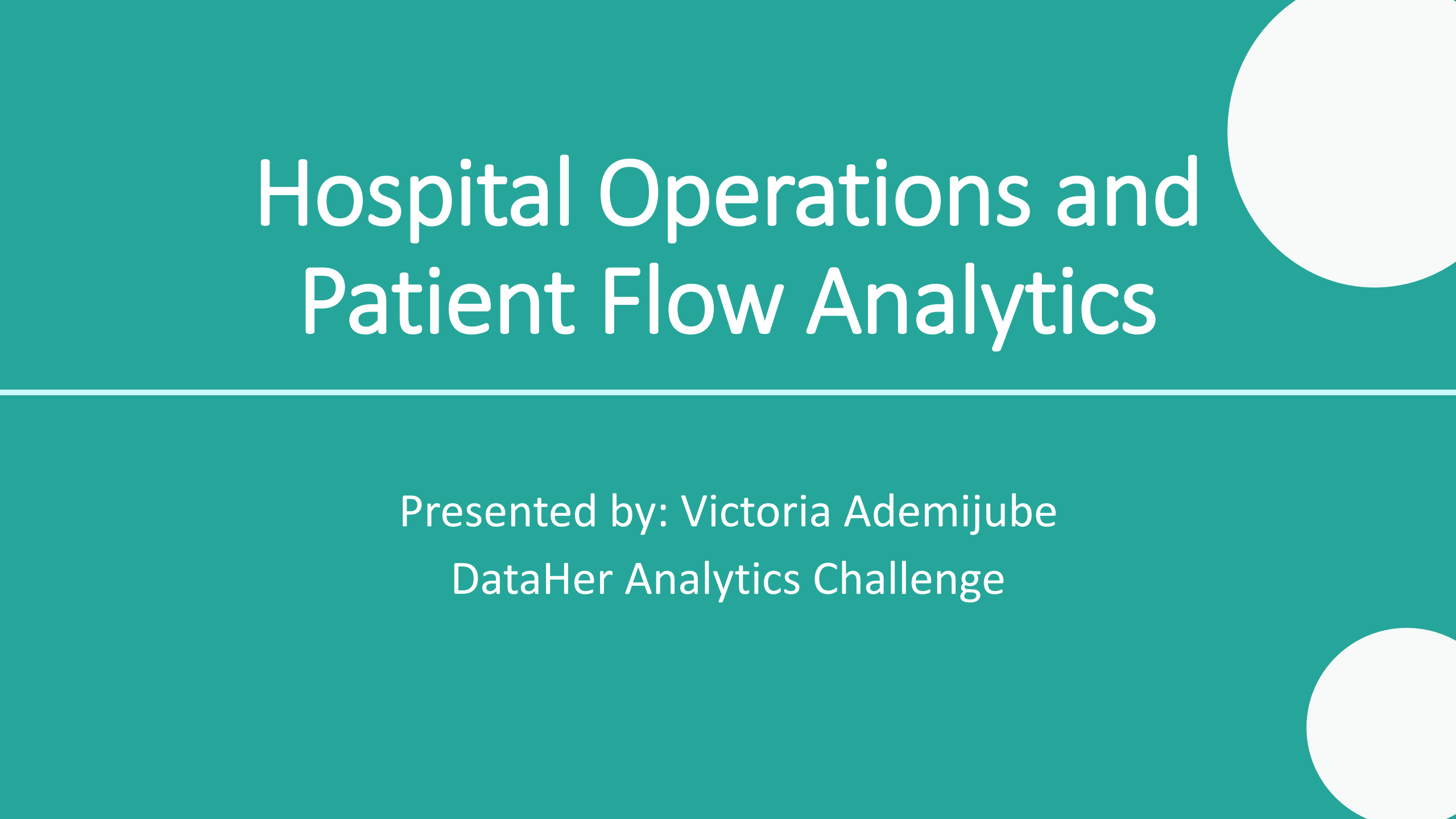


Hospital Operations and Patient Flow Analytics

The background is a solid teal color. There are two large white circles: one in the top right corner and one in the bottom right corner.

Presented by: Victoria Ademijube
DataHer Analytics Challenge

Business Questions

- Why are waiting times increasing?
- Which departments experience the longest delays?
- Which days are busiest?
- Are more doctors needed?
- Which patients wait the longest?
- Which departments receive the poorest feedback?

Methodology

Tools Used:

- Excel/Power query – data cleaning
- Power BI – data analytics/visualization
- Chat GPT- troubleshooting, brainstorming

Core KPIs

501

PATIENTS

2,000

APPOINTMENTS

25

DOCTORS

21 Mins

AVERAGE WAIT TIME

3.0

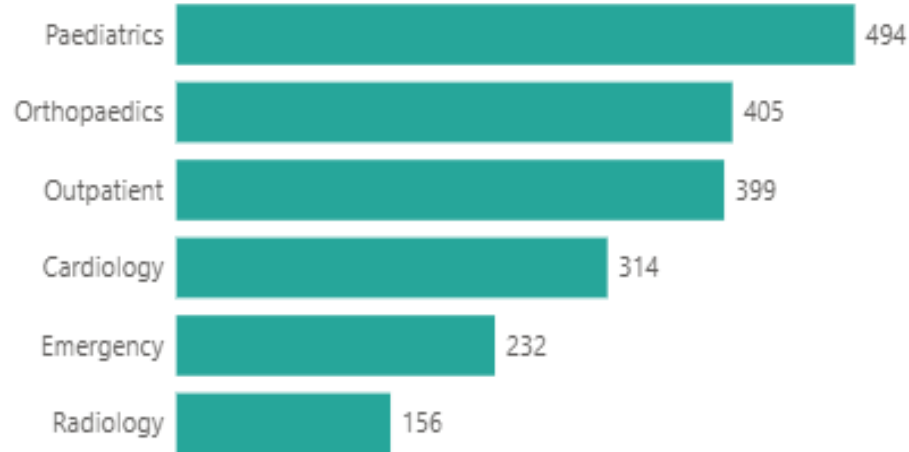
AVERAGE RATING

40 Mins

MAX WAIT TIME

Data Insights (Appointments Flow and Capacity Allocation)

Total Appiontment by Department

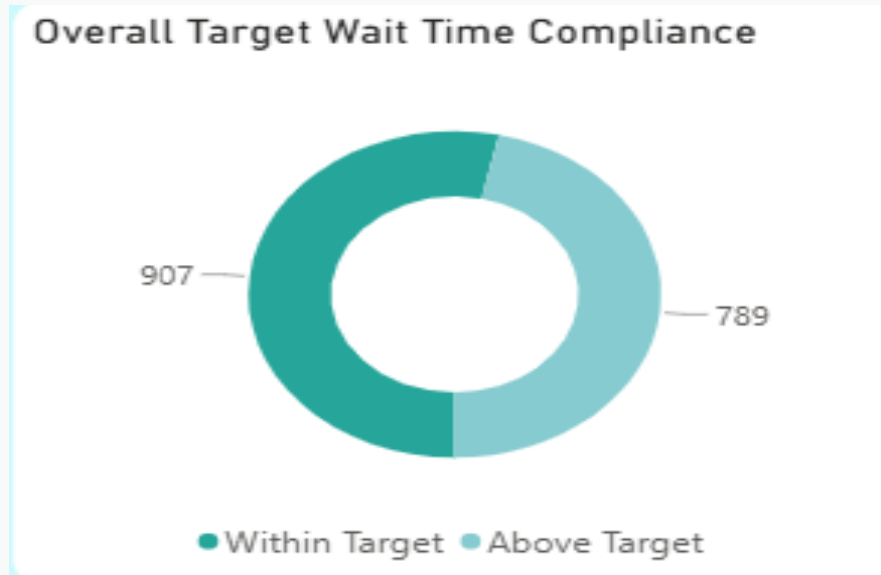


- Paediatrics recorded 494 appointments
- This is followed by Orthopaedics (405).
- These three departments account for about 65% of all appointments.

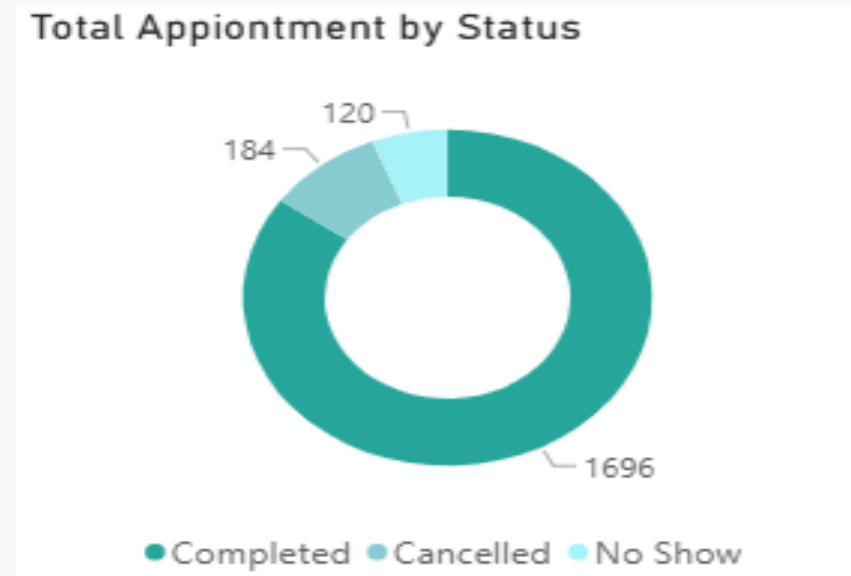
DepartmentName	Total Appiontment	Total Doctors
Paediatrics	494	6
Orthopaedics	405	5
Outpatient	399	5
Cardiology	314	4
Emergency	232	3
Radiology	156	2
Total	2000	25

- Departmental workload is relatively balanced.
- However, this is not enough to determine whether more doctors are needed.

Data Insights



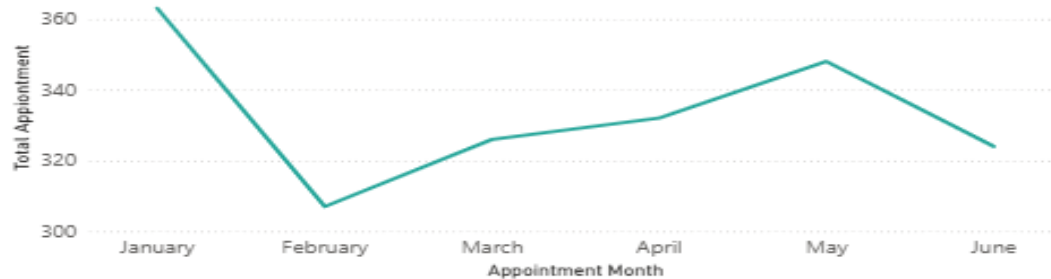
- More than 45% (789) of the completed appointments did not meet the department-specific waiting time target.
- This is still a large percentage.



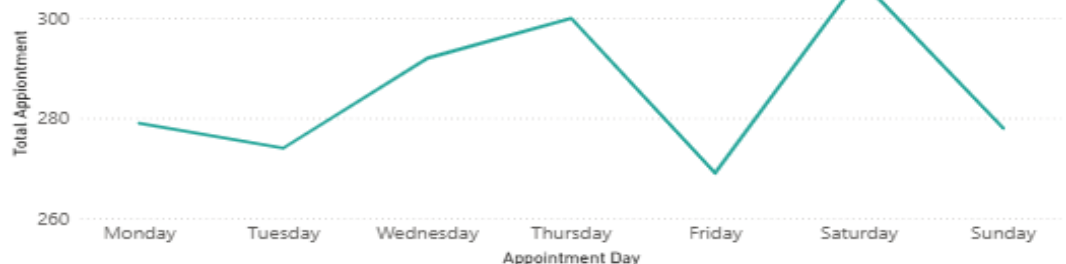
- 1,696 appointments were completed.
- 184 cancellations and 120 no-shows.

Data Insights (Monthly and Hourly patterns)

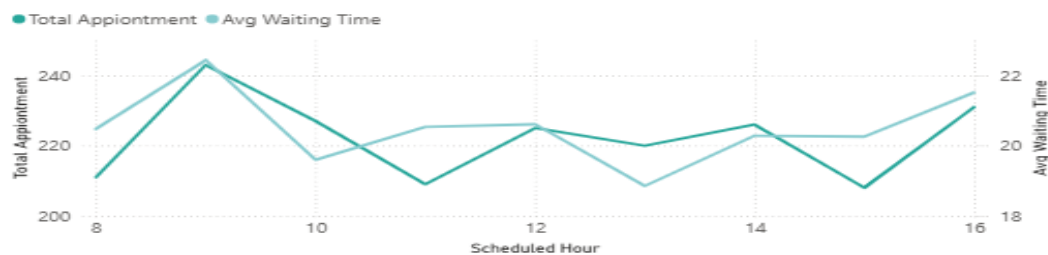
Total Appointment by Month



Total Appointment by Week Day



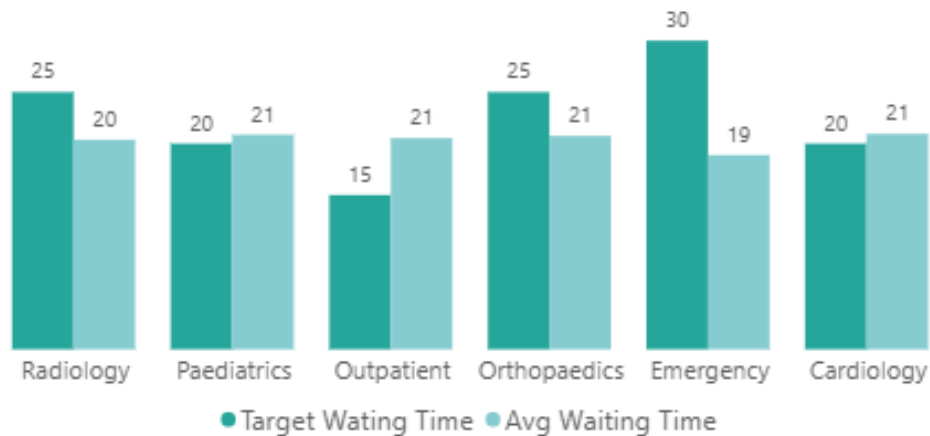
THourly Patient Volume and Avg Waiting Time



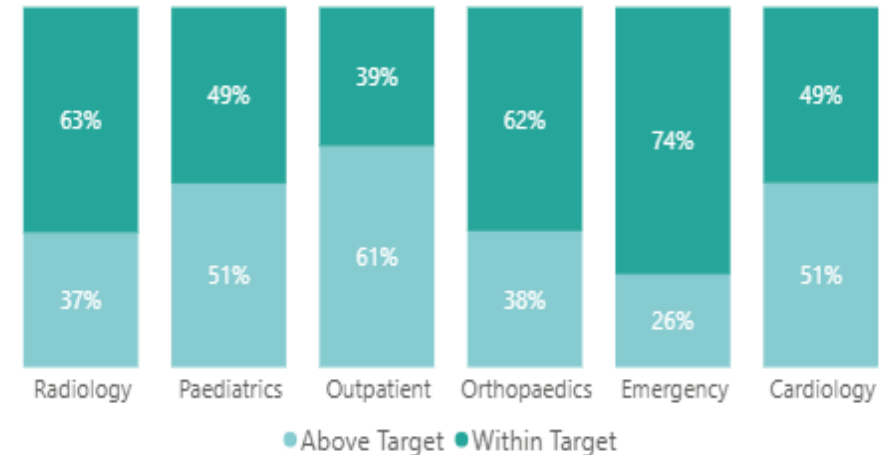
- January is the busiest month (363 appointments)
- February is the lowest (307).
- Demand rises again through May before easing in June.
- Saturday is the busiest day (308)
- Friday is the lowest (269).
- Appointment volume (240) peaks at 9am
- This is similar to average wait time at 9am. (However, the average wait time fluctuates from 19-22 mins).

Data Insights (Department-level waiting-time performance)

Target Vs Avg Wait Time by Department



Wait Time Target Compliance by Department



- Outpatients shows the greatest concern.
- Emergency has the strongest compliance with its target.
- However, its target wait time should be assessed, given the urgent nature of emergency care.
- Outpatient: highest above-target appointments (61%)
- Emergency: lowest above-target appointments (26%)

Data Insights (Doctors Workload and Availability)

Doctor	Emp Type	Appointments	Hour Worked	Consult Time(hr)
Dr E5	Part-Time	100	456	38
Dr P16	Part-Time	92	470	37
Dr N14	Part-Time	91	472	36
Dr I9	Full-Time	90	490	35
Dr W23	Full-Time	90	480	35
Dr D4	Full-Time	88	468	36
Dr M13	Part-Time	86	460	34
Dr U21	Part-Time	86	488	32
Dr C3	Part-Time	83	460	31
Dr O15	Full-Time	82	462	32
Dr V22	Part-Time	81	484	31
Dr Q17	Part-Time	80	458	31
Dr S19	Part-Time	80	496	28
Dr J10	Part-Time	78	454	31
Dr T20	Full-Time	78	476	26
Dr K11	Part-Time	76	474	34
Dr L12	Part-Time	76	476	31
Dr X24	Part-Time	76	486	30
Dr B2	Part-Time	75	474	27
Dr Y25	Part-Time	73	498	28
Dr A1	Full-Time	72	458	27
Dr G7	Part-Time	71	456	28
Dr R18	Full-Time	67	488	26
Dr F6	Part-Time	65	486	25
Dr H8	Full-Time	64	490	24
Total		2000	11860	774

- Workload is uneven
- Varying from 64 -100 appointment
- Dr E5: highest workload / part-time
- Dr H8: lowest workload/ full-time
- Scheduled capacity for each doctor ranges from 450-496 hours.
- However, recorded consultation duration was only 24-37 hours per doctor.
- Only a small portion of their scheduled working time was spent in recorded consultation.
- The remaining working hours may be spent on other clinical duties such as ward rounds.
- Long wait time may be driven by workflow factors or unrecorded activities.

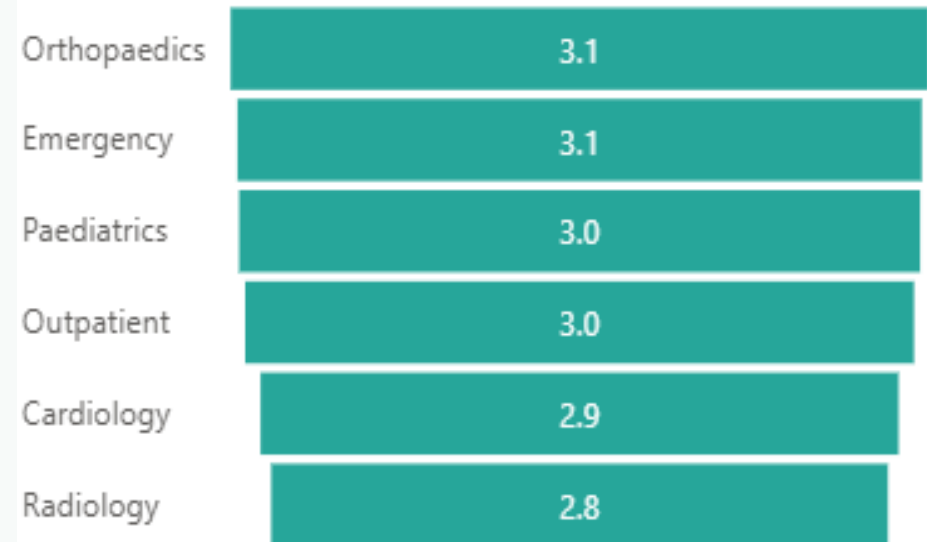
Data Insights (Patient Experience)

Patient feedback



- About 18% of the feedback suggest a need for improvement.
- 16% cited long wait time.

Average Rating by Department



- Orthopaedics and Emergency have the highest average rating at 3.1.
- Radiology is lowest at 2.8

Recommendations

- Investigate the cause of delay by conducting a process review to determine whether delays come from late clinic start, registration, or doctor availability.
- Standardize clinic workflows.
- Implement continuous performance monitoring.
- Reduce no-shows and cancellations through active reminders and better appointment-management process.
- Review appointment scheduling and capacity allocation based on doctor availability.
- Strengthen data quality controls.



Hospital Operations & Patient Flow Analytics *(Executive Summary)*

Total Patients

501

Total Appiontment

2000

Completed Appointments

1696

Complete Rate

84.80%

Avg Waiting Time (Min)

21

Executive Summary

Operations

Resources

Patient Experience

Appt Status

All

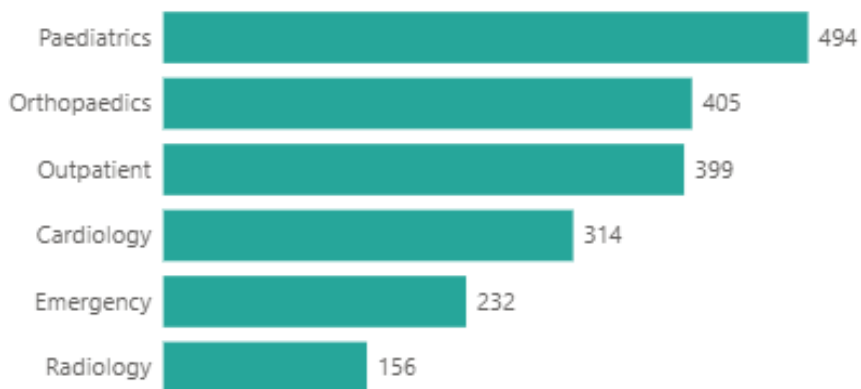


Department

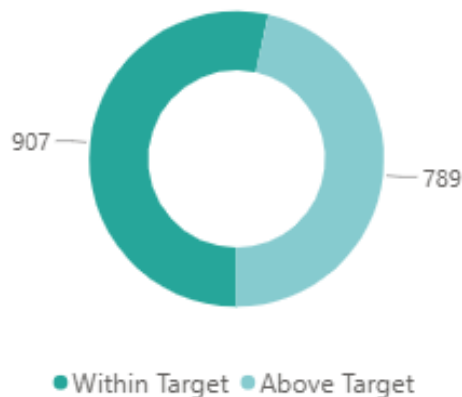
All



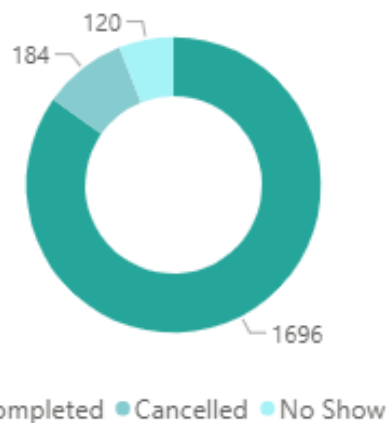
Total Appiontment by Department



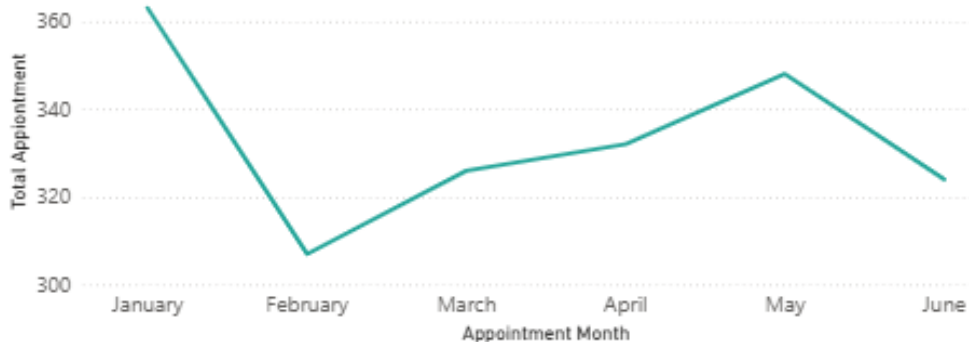
Overall Target Wait Time Compliance



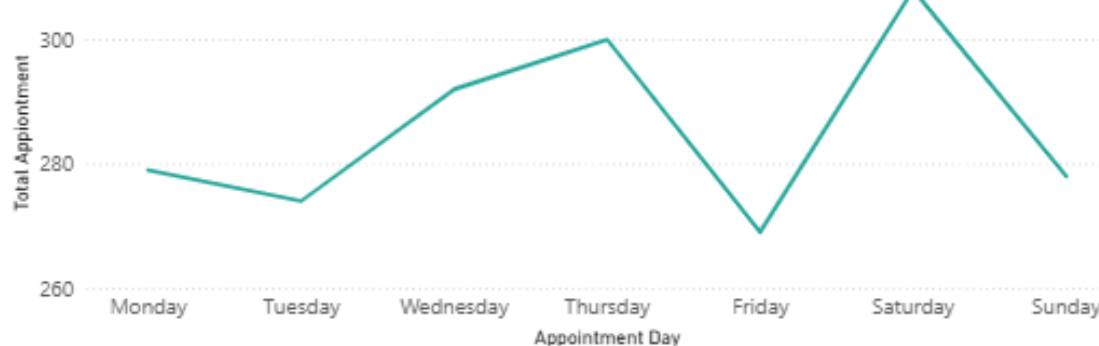
Total Appiontment by Status



Total Appointment by Month



Total Appiontment by Week Day





Hospital Operations & Patient Flow Analytics *(Operations)*

Total Patients

501

Total Appiontment

2000

Avg Waiting Time (Min)

21

Max Waiting Time (Min)

40

Within Target%

60.6%

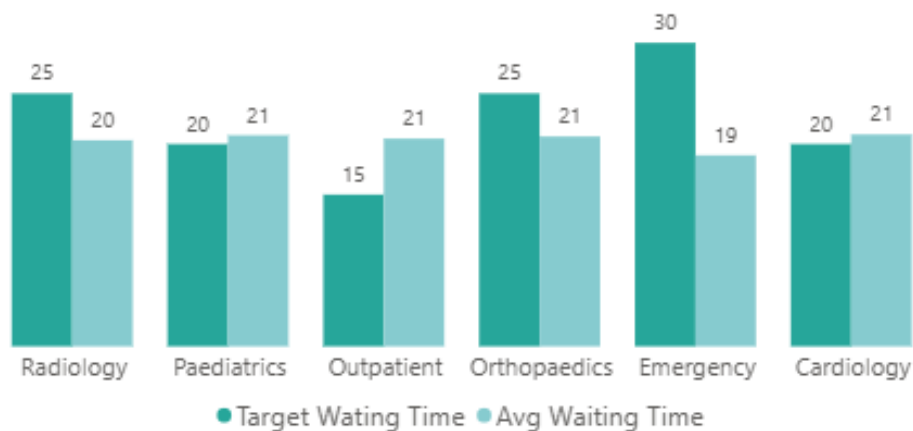
Executive Summary

Operations

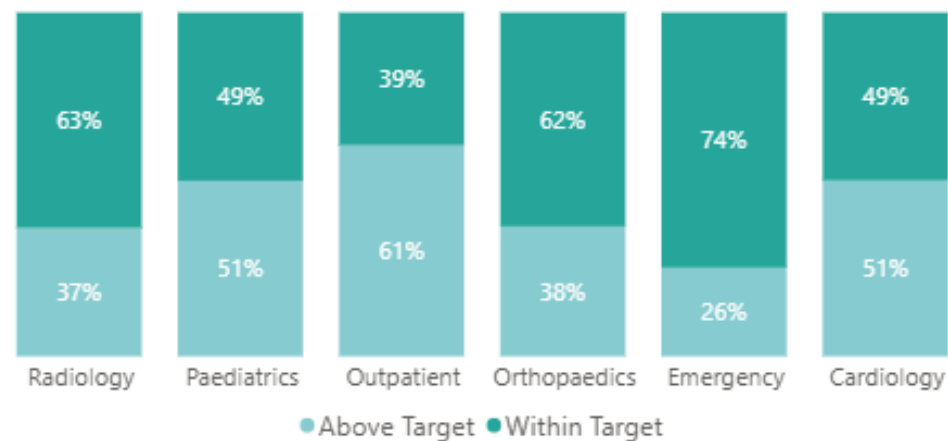
Resources

Patient Experience

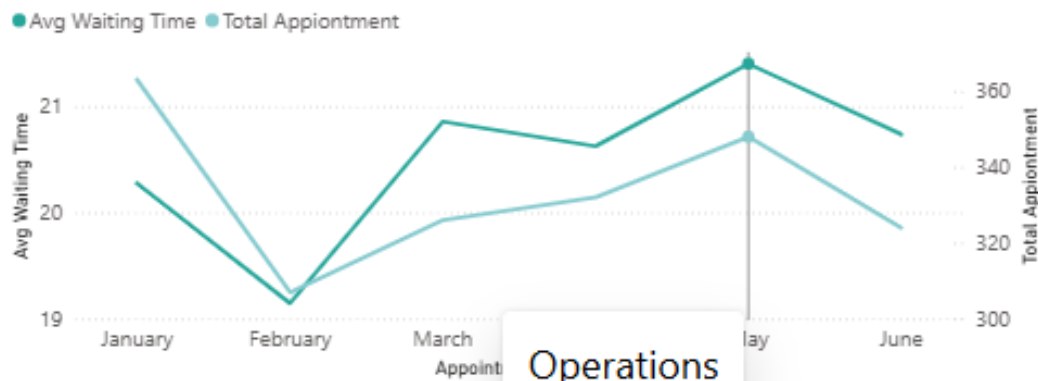
Target Vs Avg Wait Time by Department



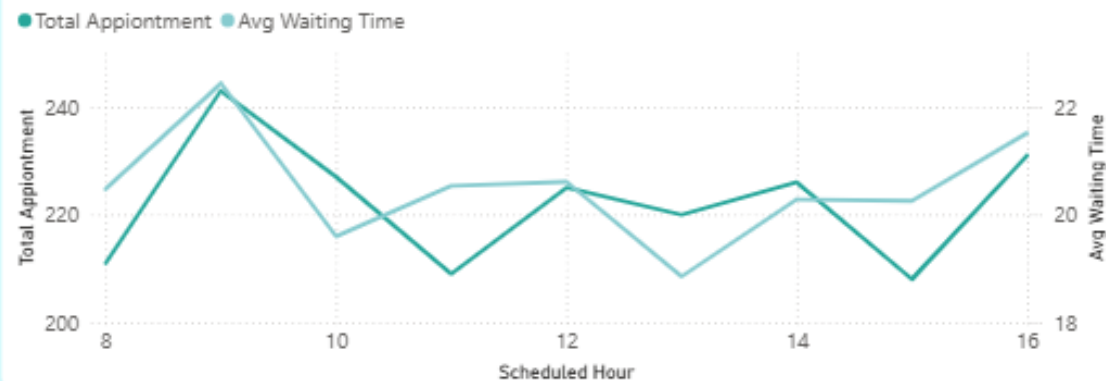
Wait Time Target Compliance by Department



Monthly Volume and Avg Wait Time Correlation



Hourly Patient Volume and Avg Waiting Time



Operations

Appt Status

All

Department

All



Hospital Operations & Patient Flow Analytics *(Resources)*

Total Patients

501

Total Doctors

25

Avg Waiting Time (Min)

21

Avg Consultation Duration
(Min)

27

Avg Consultation per Day

257

Executive Summary

Operations

Resources

Patient Experience

Appt Status

All

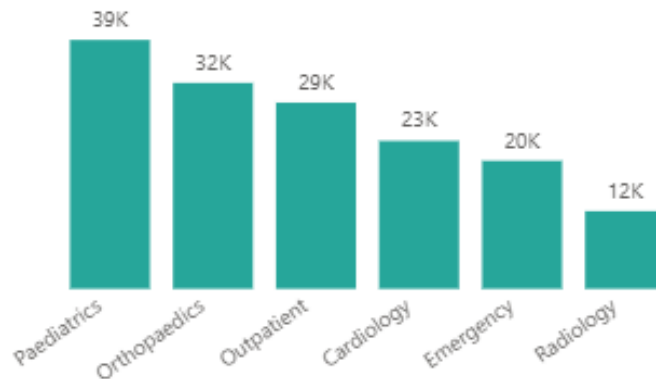
Department

All

Treatment and Procedure Volume



Treatment Cost by Department



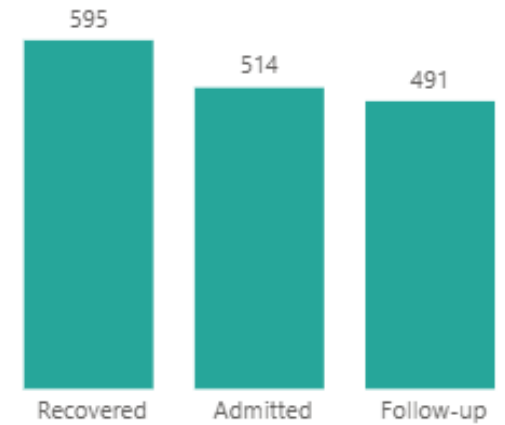
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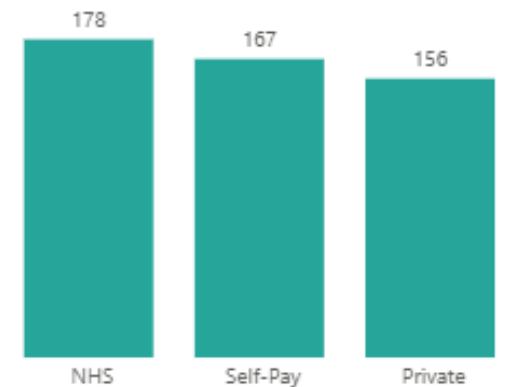
Doctor Emp Type Appointments Hour Worked Consult Time(hr)

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Dr O15	Full-Time	82	462	32
Dr V22	Part-Time	81	484	31
Dr Q17	Part-Time	80	458	31
Total		2000	11860	774

Patient Care Outcome



Total Patients by InsuranceType





Hospital Operations & Patient Flow Analytics *(Patient Experience)*

Total Patients

501

Total Appiointment

2000

Avg Waiting Time (Min)

21

Average Rating

3.0

Recommendation Rate

50.3%

Executive Summary

Operations

Resources

Patient Experience

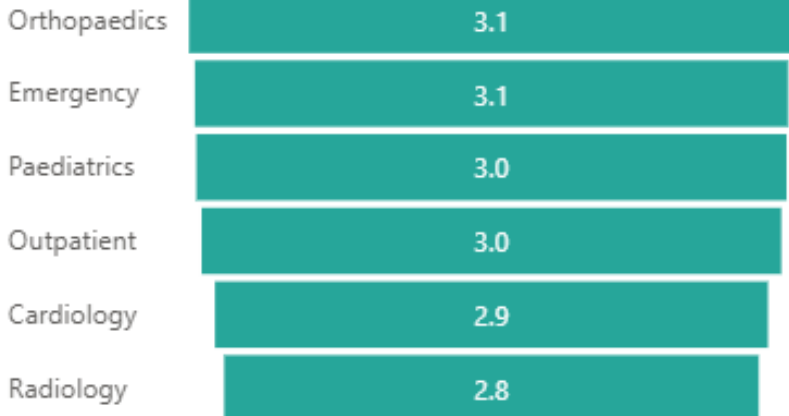
Patient Hospital Recommendation Rate



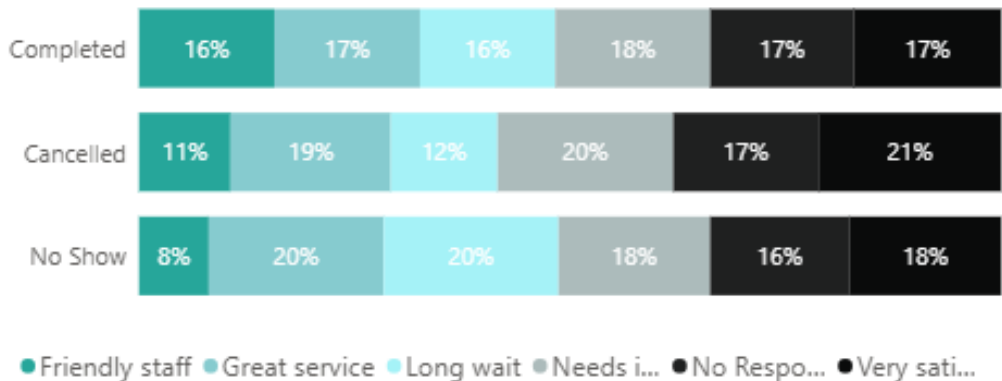
Patient feedback



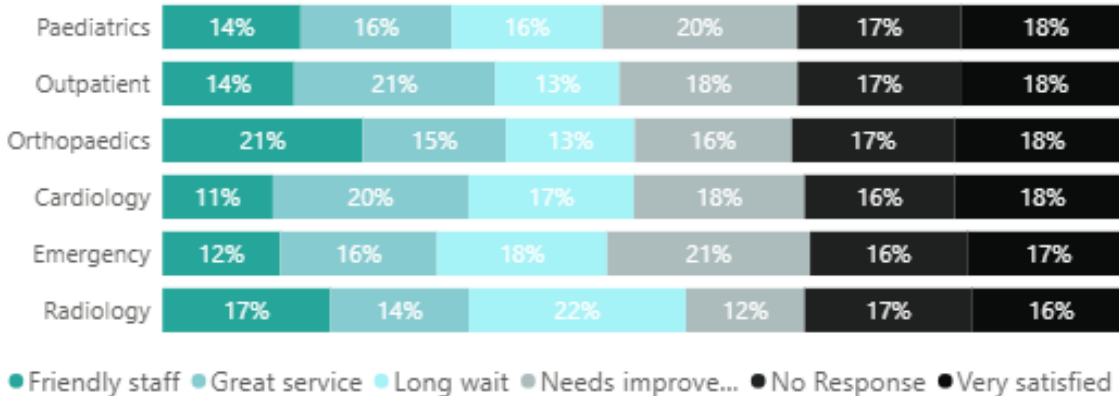
Average Rating by Department



Feedback Category by Appointment Status



Feedback Category by Department



Appointment Stat...

All

Department

All